# National University of Computer and Emerging Sciences

# Phase 1 Project Proposal: NeuroScan

## **Group ID:** F26-07

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**Track:** Track A (Application Development)

# 1. Problem Statement & Solution

Manual interpretation of brain MRI scans is time-consuming and prone to human error, especially in high-pressure clinical environments. Distinguishing a tumor from healthy brain tissue is visually complex. To assist with this, patients are often injected with a metallic contrast dye called **Gadolinium** prior to the MRI. Because brain tumors grow rapidly, they create poorly formed, "leaky" blood vessels. The Gadolinium dye leaks out of these vessels and pools directly inside the tumor tissue. When exposed to the MRI's magnetic field, the pooled Gadolinium causes the tumor to light up brightly against the dark background of healthy tissue.

Despite this chemical enhancement, analyzing hundreds of slices per scan remains a massive cognitive load for doctors. Furthermore, once a diagnosis is made, **communicating these complex visual findings to surgical teams or patients is currently inefficient and often lacks standardized, secure documentation**.

**NeuroScan solves these problems** by providing an AI-powered diagnostic assistant that utilizes Deep Learning (specifically Convolutional Neural Networks) to detect and localize intracranial tumors. By automatically detecting the tumor and highlighting its exact location, the goal is to provide a "second opinion" for radiologists to improve diagnostic speed and accuracy. To solve the communication bottleneck, NeuroScan bridges the gap by automatically generating secure, standardized PDF reports of these visual findings, ensuring seamless and rapid information transfer between medical departments.

# 2. Dataset

- **Primary Dataset:** Brain Tumor MRI Dataset
- **Dataset Link:** [Kaggle: Brain Tumor MRI Dataset]
- **Description:** These datasets contain T1-weighted contrast-enhanced images with three types of tumors: Meningioma, Glioma, and Pituitary tumors.

## 3. Scope of the Project

To meet the requirements of Track A, the project will encompass:

- **Backend Logic:** Development of a CNN model (e.g., ResNet or EfficientNet) to classify and segment tumor regions.
- **Preprocessing:** Normalization, skull-stripping, and data augmentation to handle various MRI intensities.
- **Live Visual User Interface:** A web-based dashboard (using Flask) where medical professionals can upload MRI scans. **Crucially, before any reports are generated, the dashboard will immediately display the original scan alongside a localized heat map (Grad-CAM) showing the detected tumor area directly on the screen for the doctor to review**.
- **Automated PDF Report Generation (Extended Scope):** Once the doctor reviews the on-screen heat map, they can click a button to trigger a Flask routing system with Jinja2 templating. This will dynamically inject the patient data, the AI's classification results, and the live Grad-CAM heat map into a standard HTML layout, automatically generating a highly professional, downloadable diagnostic PDF report.
- **Secure Document Sharing (Extended Scope):** Development of a database-backed sharing architecture using Python's `uuid` library. This will allow the system to generate secure, randomized, and temporary short-links for the generated PDF reports, enabling fast and confidential sharing between radiologists and surgical teams.
- **Deployment:** Hosting the application for real-time inference demonstrations.

## 4. Relevant Research Paper

- **Paper Title:** Explainable CNN for brain tumor detection and classification through XAI based key features identification
- **IEEE Citation:** S. Iftikhar, N. Anjum, A. B. Siddiqui, M. Ur Rehman, and N. Ramzan, "Explainable CNN for brain tumor detection and classification through XAI based key features identification," *Brain Informatics*, vol. 12, no. 1, p. 10, Apr. 2025.
- **Paper Links:** * [SpringerOpen Official Publication]

    - PMC PubMed Central Full Text
- **Relevance:** This paper directly mirrors the NeuroScan project architecture. It validates the methodology of using Convolutional Neural Networks to classify Glioma, Meningioma, and Pituitary tumors from MRI datasets. Additionally, it strongly supports the use of Explainable AI (XAI) tools like Grad-CAM to generate heat maps, proving that this visual approach makes the AI's diagnostic decisions more transparent and trustworthy for medical professionals.